

Problem 6.15

Fund A is invested at an effective annual interest rate of 3%. Fund B is invested at an effective annual interest rate of 2.5%. At the end of 20 years, the total in the two funds is 10,000. At the end of 31 years, the amount in Fund A is twice the amount in Fund B . Calculate the total in the two funds at the end of 10 years.

Let the initial amounts invested in Funds A and B be A and B , respectively.

Then at the end of 20 years,

$$A(1.03)^{20} + B(1.025)^{20} = 10000.$$

At the end of 31 years, the amount in Fund A is twice the amount in Fund B , so

$$A(1.03)^{31} = 2B(1.025)^{31}.$$

Hence,

$$A = 2B \left(\frac{1.025}{1.03} \right)^{31}.$$

Substituting this into the 20-year equation gives

$$2B \left(\frac{1.025}{1.03} \right)^{31} (1.03)^{20} + B(1.025)^{20} = 10000.$$

So,

$$2B \frac{(1.025)^{31}}{(1.03)^{11}} + B(1.025)^{20} = 10000.$$

Solving, we obtain

$$B \approx 2107.46.$$

Then

$$A = 2B \left(\frac{1.025}{1.03} \right)^{31} \approx 3624.13.$$

Now compute the total at the end of 10 years:

$$A(1.03)^{10} + B(1.025)^{10}.$$

Substituting the values of A and B ,

$$3624.13(1.03)^{10} + 2107.46(1.025)^{10} \approx 4871.33 + 2697.73.$$

Therefore, the total at the end of 10 years is

$$\boxed{7569.06}.$$